Preview: y2007_q03 (v3)

Comparing sets of letters: each different letter counts once; check them all

Part A: Each question is worth 3 points

1. How many letters do the words ELEPHANT and NUMBER have in common?

A. 4 ***B.*** 3 ***C.*** 2 ***D.*** 1 ***E.*** 0

2. How many letters do the words ELEPHANT and SPIDER have in common?

A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

3. How many different letters are used to write the word CHOCOLATE?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

4. How many different letters are used to write the word WINDOW?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

Part B: Each question is worth 4 points

5. Mia writes the word MATHEMATICS. Then Mia crosses out every letter that also appears in the word TRIANGLE.
How many different letters are not crossed out?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

6. Leo writes the word SCHOOL. Then Leo crosses out every letter that also appears in the word KANGAROO.
How many different letters are not crossed out?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

7. Which word has exactly 3 different letters in common with PENCIL?

A. DRAGON ***B.*** MONKEY ***C.*** SPIDER ***D.*** TEACHER ***E.*** KANGAROO

8. Which word has exactly 2 different letters in common with CHOCOLATE?

A. PROBLEM ***B.*** PLANET ***C.*** CASTLE ***D.*** SQUARE ***E.*** SPIDER

Part C: Each question is worth 5 points

9. How many different letters are used to write the two words MATHEMATICS and PLANET?

A. 9 ***B.*** 10 ***C.*** 11 ***D.*** 12 ***E.*** 17

10. How many different letters are used to write the two words KANGAROO and RAINBOW?

A. 7 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 13

Answer Key

1. **C**	**2. E**	**3. B**	**4. A**	**5. E**	**6. B**
7. **C**	**8. D**	**9. C**	**10. C**		

Solutions

1. C (2) — Check each different letter of ELEPHANT: only E and N are also in NUMBER, so 2. A letter that appears twice still counts once. [2007 Q3 · common_letters]

2. E (2) — Check each different letter of ELEPHANT: only E and P are also in SPIDER, so 2. A letter that appears twice still counts once. [2007 Q3 · common_letters]

3. B (7) — The letters are A, C, E, H, L, O and T: 7 different letters (CHOCOLATE has 9 letters in all). [2007 Q3 · different_letters]

4. A (5) — The letters are D, I, N, O and W: 5 different letters (WINDOW has 6 letters in all). [2007 Q3 · different_letters]

5. E (4) — Crossed out: A, E, I and T (they are in TRIANGLE). Left: C, H, M and S, so 4 different letters (a letter that is left twice counts once). [2007 Q3 · only_in_first]

6. B (4) — Crossed out: O (they are in KANGAROO). Left: C, H, L and S, so 4 different letters. [2007 Q3 · only_in_first]

7. C (SPIDER) — SPIDER shares E, I and P with PENCIL. KANGAROO shares 1; DRAGON shares 1; MONKEY shares 2; TEACHER shares 2. [2007 Q3 · which_word]

8. D (SQUARE) — SQUARE shares A and E with CHOCOLATE. PROBLEM shares 3; PLANET shares 4; SPIDER shares 1; CASTLE shares 5. [2007 Q3 · which_word]

9. C (11) — MATHEMATICS uses 8 different letters and PLANET uses 6. A, E and T are in both, so count them once: 8 + 6 – 3 = 11. [2007 Q3 · letter_cards]

10. C (9) — KANGAROO uses 6 different letters and RAINBOW uses 7. A, N, O and R are in both, so count them once: 6 + 7 – 4 = 9. [2007 Q3 · letter_cards]